

Sri Lanka Institute of Information Technology



SE2030 – Software Engineering
Year 2, Semester 1- 2026

Web-based Diet Planning & Health Check-up System

Lab 01 Report

2026-Y2-S1-MLB-B5G1-01

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Summary of requirement gathering process

Hi! I am developing a Web-based Diet Planning & Health Check-up System for my software engineering project. We want to build a platform that connects patients with dietitians and handles medical appointments. Can you help me gather requirements?



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Thank you for the project details.

Our client company is **Lanka Wellness Care (Pvt) Ltd**, a private healthcare and nutrition service provider in Sri Lanka. They want a new project called **NutriCare Connect**, a web-based system to help patients, dietitians, and medical staff work together. The system should support diet plans, health check-ups, appointment management, and patient follow-up.

Here are the client representatives you can talk to:

Mr. Nuwan Perera – Operations Manager

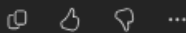
Ms. Ishara Jayasinghe – Senior Dietitian

Mr. Chamara Fernando – Medical Center Coordinator

Ms. Dilki Ranatunga – Patient Relations Officer

Mr. Sahan Wickramasinghe – Finance and Administration Executive

You can now ask any question about the project requirements, and the most suitable person will answer as a real client.



Hello Mr. Nuwan. To help us understand the core purpose of NutriCare Connect, what are the biggest operational bottlenecks your team currently faces, and what specific workflows must this system streamline first?



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Mr. Nuwan Perera – Operations Manager

That is a very good question.

At the moment, our biggest problem is that information is scattered. Patient records, appointment details, diet plans, and follow-up notes are often kept in different files or systems. Staff spend a lot of time searching for information and updating records manually.

The first workflow we want to streamline is the **patient journey from registration to follow-up**. A patient should be able to register, book an appointment, attend a consultation, receive a diet plan, and track progress through one platform.

Another major issue is appointment management. We sometimes face double bookings, missed appointments, and difficulty coordinating schedules between dietitians, doctors, and patients. The new system should make scheduling much easier.

We also need better patient progress tracking. Dietitians should be able to see previous consultations, health measurements, and diet plans without searching through multiple records.

From an operations perspective, the highest-priority workflows are:

1. Patient registration and profile management.
2. Appointment booking, approval, and reminders.
3. Health check-up record management.
4. Diet plan creation and sharing with patients.
5. Follow-up consultation and progress monitoring.
6. Basic billing and payment record tracking.

Success for us means less paperwork, fewer missed appointments, faster service for patients, and better visibility of patient health progress.

Feel free to ask more detailed questions. For example, you can ask: "Can you describe how a patient currently books an appointment and how you want that process to work in the future?"



Can you describe how a patient currently books an appointment and how you want that process to work in the future?

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Mr. Nuwan Perera – Operations Manager

Currently, most patients book appointments by calling our reception desk, sending a message, or visiting the center in person. Reception staff then check availability manually and record the appointment in spreadsheets or appointment books. This process takes time and sometimes leads to scheduling mistakes or missed updates.

In the future, we would like patients to create an account and book appointments online. They should be able to view available dates and time slots for dietitians and doctors, choose a suitable time, and receive immediate confirmation.

We also want the system to send automatic reminders before the appointment. If a patient needs to cancel or reschedule, they should be able to do it through the system without calling the center.

For our staff, the system should provide a central appointment calendar so everyone can see schedules in real time. This will help us avoid double bookings and reduce administrative work.

A typical future process would be:

A patient registers → views available specialists → selects a time slot → receives confirmation → gets reminder notifications → attends the consultation → follow-up appointment is scheduled if needed.

This workflow is one of our highest priorities because appointment management affects almost every department in our organization.

Hi Ms. Ishara. When you create and assign a diet plan, what specific data such as calorie counts, allergens, or meal schedules must the system capture? Additionally, how do you want the system to help you track if a patient is actually following the plan?



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Ms. Ishara Jayasinghe – Senior Dietitian

That is an important requirement for us.

When I create a diet plan, I need the system to record the patient's basic health information such as age, gender, height, weight, BMI, medical conditions, allergies, food restrictions, medications, and lifestyle information. I also need details such as target weight, daily calorie goals, and nutritional goals.

For the diet plan itself, I would like to record meal schedules, recommended foods, foods to avoid, portion sizes, daily water intake, and any special instructions. Some patients may also need diet plans for conditions such as diabetes, high blood pressure, or obesity, so the system should support condition-specific recommendations.

To track whether a patient is following the plan, I would like patients to regularly update their weight and basic health measurements through the system. It would also be helpful if patients could mark meals as completed and keep a simple food diary.

During follow-up consultations, I would like to compare the patient's actual progress with the expected goals. For example, I should be able to see whether their weight is changing as planned, whether they are attending follow-up appointments, and whether they have been recording their meals consistently.

A dashboard showing progress trends would be very useful. If a patient stops updating information or misses follow-ups, the system could alert the dietitian so that we can contact the patient and provide support.

From my perspective, a successful system is not only one that stores diet plans but also helps us understand whether patients are achieving their nutrition and health goals.

If you are collecting requirements, you could also ask me: "How often do diet plans need to be reviewed or updated, and who should be allowed to make those changes?" This may help identify important functional requirements and user permissions.

What reports or statistics would management and medical staff like to see from the system?

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Mr. Chamara Fernando – Medical Center Coordinator

That is a very important area for us. We need reports that help both management and medical staff make decisions.

For management, we would like to see reports such as:

- Number of appointments booked, completed, cancelled, and missed.
- Daily, weekly, and monthly patient visits.
- Most popular health check-up packages and services.
- Doctor and dietitian workload statistics.
- Revenue, payments received, and outstanding payments.
- Patient registration growth over time.
- Patient satisfaction or feedback summaries.
- Follow-up completion rates.

These reports help management understand how well the center is performing and identify areas that need improvement.

For medical staff, we would like reports such as:

- Patient health history and consultation records.
- Health measurement trends (weight, BMI, blood pressure, blood sugar, etc.).
- Progress against diet and health goals.
- Upcoming follow-up appointments.
- Patients who missed appointments or stopped follow-ups.
- Patients requiring urgent attention based on health indicators.
- Diet plan effectiveness and patient compliance summaries.

We would also like dashboards with charts and graphs because they are easier to understand than large tables of data.

Another useful feature would be the ability to filter reports by date range, doctor, dietitian, health condition, or check-up package.

From a business perspective, one of the most valuable reports would be identifying how many patients successfully complete their treatment or nutrition programs, as this helps us measure the quality of our services.

What are the information that should be recorded during a health check-up, and who should have access to those records?

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Mr. Chamara Fernando – Medical Center Coordinator

Yes, certainly.

During a health check-up, we would like the system to record the patient's personal details, appointment details, and all medical observations taken during the visit.

The information may include the patient's height, weight, BMI, blood pressure, blood sugar levels, heart rate, temperature, allergies, existing medical conditions, medications, laboratory test results, and the doctor's or dietitian's notes. We also need the date of the check-up, the staff member who performed it, and any recommendations given to the patient.

If follow-up actions are required, such as a diet consultation, additional tests, or another medical appointment, those details should also be recorded.

Regarding access, different users should have different permissions.

Patients should be able to view their own health records, test results, recommendations, and appointment history.

Doctors and dietitians should be able to access the medical records of patients under their care so they can make informed decisions.

Reception staff should only see information needed for appointment management and patient registration. They should not have access to sensitive medical details unless authorized.

System administrators may need access for management and auditing purposes, but access to medical records should be controlled and logged for privacy reasons.

Patient health information is confidential, so the system should ensure that only authorized users can view or update records.

This is very important for us because accurate and secure medical records help ensure better patient care and follow-up services.

Hello Mr. Chamara. Could you walk me through the ideal workflow for a patient booking a health check-up? Specifically, how should the system handle conflicts like double bookings, doctor unavailability, or last-minute cancellations?

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Mr. Chamara Fernando – Medical Center Coordinator

Hello. I can explain how we would like the health check-up booking process to work.

In the ideal process, a patient logs into the system and selects a health check-up package or a specific doctor. The system should show only available dates and time slots. Once the patient chooses a slot, the booking should be confirmed immediately and added to the doctor's schedule.

To prevent **double bookings**, the system should automatically block a time slot as soon as it is reserved. Two patients should never be able to book the same slot.

If a doctor becomes **unavailable** because of leave, an emergency, or schedule changes, the system should notify affected patients immediately. Patients should be offered alternative dates, times, or another suitable doctor if available.

For **last-minute cancellations**, the cancelled slot should become available for other patients right away. Patients on a waiting list could be notified that a slot has opened. This helps us make better use of available appointment times.

We would also like automatic reminders before appointments. If a patient does not confirm attendance or misses appointments repeatedly, staff should be able to see this information.

A typical workflow would be:

Patient selects check-up → System shows available slots → Patient confirms booking → System sends confirmation and reminders → Patient attends check-up → Results and follow-up appointment are recorded if needed.

One important requirement is that reception staff should also be able to create, modify, or cancel bookings on behalf of patients who call or visit the center.

What types of notifications, reminders, or feedback features would patients find most valuable?

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Ms. Dilki Ranatunga – Patient Relations Officer

That is a very good question.

Patients often tell us that they miss appointments or forget health instructions because they are busy. So, notifications and reminders are very important.

The most valuable reminders would be appointment reminders sent a few days before the visit and again on the appointment day. Patients should also receive notifications when an appointment is booked, rescheduled, or cancelled.

For patients following diet plans, reminders for meal schedules, water intake, medication, and follow-up consultations would be very useful. We would also like the system to notify patients when their dietitian updates a diet plan or adds new recommendations.

Regarding feedback features, patients should be able to rate their consultation experience and provide comments after appointments. This helps us understand service quality and identify areas for improvement.

Another useful feature would be simple health progress notifications. For example, the system could congratulate patients when they achieve a weight-loss goal or remind them when they have not updated their health information for some time.

From the patient's point of view, the most useful notifications are:

- Appointment confirmations and reminders.
- Follow-up appointment reminders.
- Diet plan update alerts.
- Medication and meal reminders.
- Health record or test result availability notifications.
- Payment due reminders.
- Health goal achievement notifications.

The feedback features we would most like to see are simple satisfaction ratings, comments, and the ability for patients to report issues or request assistance through the portal.

These features would help patients stay engaged with their healthcare journey and reduce the number of missed appointments and follow-ups.

Hello Mr. Sahan. Regarding the administrative and financial side of Lanka Wellness Care, what specific types of data or activity summaries do you need the system to compile into monthly reports?

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Mr. Sahan Wickramasinghe – Finance and Administration Executive

Hello. From the administration and finance side, monthly reports are very important for monitoring the business.

We would like the system to prepare reports on revenue and payments. This should include total income for the month, income by service type (diet consultations, health check-ups, follow-up sessions, etc.), pending payments, completed payments, and refunded payments.

We also need patient activity summaries. These should show:

- New patient registrations.
- Total active patients.
- Number of appointments booked, completed, cancelled, and missed.
- Number of follow-up consultations conducted.

For resource planning, we would like to see:

- Doctor and dietitian consultation counts.
- Staff workload summaries.
- Room or facility utilization statistics.

Management would also benefit from reports showing:

- Most frequently requested health services.
- Most common health conditions being treated.
- Patient retention rates (patients returning for follow-ups).
- Patient feedback and satisfaction ratings.

For financial audits, the system should keep:

- Payment transaction history.
- Invoice records.
- User activity logs related to financial transactions.
- Report export history.

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Ideally, management should be able to generate these reports for different periods such as daily, weekly, monthly, quarterly, and yearly.

One report that is especially valuable for us is a **Monthly Business Performance Summary** that combines:

- Total patients served.
- Total appointments.
- Revenue generated.
- Outstanding payments.
- Service popularity.
- Patient satisfaction score.

This gives management a quick overview of the organization's performance.

Hi Ms. Dilki. From a patient's perspective, what are the current frustrations when trying to communicate with dietitians or access medical advice, and what specific features would make this new portal more user-friendly for them?

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Ms. Dilki Ranatunga – Patient Relations Officer

Hello. From the patient's side, there are a few common frustrations we hear about.

Many patients find it difficult to contact the correct person when they have questions. Sometimes they have to make multiple phone calls to get updates about appointments, diet plans, or test results. Patients also feel frustrated when they need to visit the center just to access information that could be available online.

Another challenge is that patients may forget appointment dates, follow-up schedules, or diet instructions. Some patients also feel uncomfortable asking questions after a consultation because there is no easy communication channel.

For the new NutriCare Connect system, I would like patients to have a simple and user-friendly portal where they can:

- View upcoming and past appointments.
- Book, cancel, or reschedule appointments online.
- Access their diet plans and health records anytime.
- View check-up results and recommendations.
- Receive appointment and medication reminders.
- Track their health progress with simple charts.
- Receive notifications when a diet plan is updated.
- Submit questions to their dietitian or medical staff.
- View payment history and invoices.

We should also remember that not all patients are highly technical. The system should have a clean design, simple language, and easy navigation.

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We should also remember that not all patients are highly technical. The system should have a clean design, simple language, and easy navigation.

One feature that would greatly improve the patient experience is a secure messaging system where patients can send non-urgent questions to their dietitian and receive responses without needing to call the center.

Another useful feature would be a personalized dashboard that immediately shows:

- Next appointment
- Current diet plan
- Health progress summary
- Pending payments
- Important reminders

From a patient relations perspective, success means that patients feel informed, supported, and able to manage their healthcare without unnecessary effort.

What payment methods should the system support, and what information must be stored for each transaction?

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Mr. Sahan Wickramasinghe – Finance and Administration Executive

Hello. Yes, I can explain that.

We would like the system to support several payment methods because different patients prefer different options. The most important payment methods are:

- Cash payments at the reception desk.
- Debit card payments.
- Credit card payments.
- Online bank transfers.
- Online payment gateway payments.
- Mobile wallet or digital payment options (if available).

For each transaction, the system should store enough information for financial tracking and audits. We would typically need:

- Transaction ID or reference number.
- Invoice number.
- Patient name and patient ID.
- Service purchased (diet consultation, health check-up, follow-up visit, etc.).
- Amount charged.
- Date and time of payment.
- Payment method used.
- Payment status (paid, pending, partially paid, refunded).
- Staff member who processed the payment (for in-person payments).
- Notes or remarks if necessary.
- Refund details, if a refund is issued.

We would also like the system to automatically generate invoices and keep a history of all transactions. Patients should be able to view their payment history, while finance staff should be able to search, filter, and export payment records.

Another important requirement is that every financial action should be logged. If a payment is modified, cancelled, or refunded, the system should record who made the change and when it happened.

From the finance department's perspective, the goals are:

- Accurate payment tracking.
- Easy invoice management.
- Reduced manual bookkeeping.
- Better financial reporting.
- Transparent audit trails.

This information will help you identify payment management requirements, invoice requirements, and audit/logging requirements for your Software Engineering project.

Ms. Ishara , How should the system connect a patient's health check-up results with their dietary management? For example, what kind of automated notifications or alerts should a dietitian receive when a patient's new medical test results are uploaded?

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Ms. Ishara Jayasinghe – Senior Dietitian

That is an excellent question because health check-up results and diet management should work closely together.

When new medical test results are uploaded, I would like the system to automatically link them to the patient's profile and compare them with previous results. This helps me quickly identify important changes without manually reviewing every record.

For example, I would like to receive alerts when:

- A patient's blood sugar level is above the target range.
- Blood pressure readings become unusually high.
- Cholesterol levels increase significantly.
- Weight changes rapidly within a short period.
- BMI moves into a higher-risk category.
- Laboratory results indicate possible nutrition-related concerns.
- A patient is not progressing toward agreed health goals.

The system should not only notify me that new results are available but also highlight what has changed since the previous check-up.

For example, instead of showing only: *"New test results uploaded"*

I would prefer: *"Patient's blood sugar increased from 110 mg/dL to 145 mg/dL since the previous check-up."*

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I would prefer: *"Patient's blood sugar increased from 110 mg/dL to 145 mg/dL since the previous check-up."*

This helps me decide whether a diet plan review is needed.

After receiving an alert, I would like the system to provide options such as:

- Review the patient's current diet plan.
- Schedule a follow-up consultation.
- Send a message to the patient.
- Record new dietary recommendations.

The system could also categorize alerts by priority:

- **High Priority:** Critical health changes requiring immediate attention.
- **Medium Priority:** Noticeable changes that may require a diet review.
- **Low Priority:** Minor changes for routine monitoring.

Another useful feature would be a patient progress dashboard where I can see trends for:

- Weight
- BMI
- Blood sugar
- Blood pressure
- Cholesterol
- Target goals

This would help me understand whether the current diet plan is working.

From a requirement perspective, I would expect the system to automatically support actions such as:

New Check-up Result Uploaded → Compare with Previous Results → Generate Alert if Threshold is Exceeded → Notify Dietitian → Dietitian Reviews Results → Update Diet Plan or Schedule Follow-up

This integration would help us provide faster and more personalized nutritional care instead of waiting until the next appointment to discover important health changes.

As a requirement-gathering note, this suggests functional requirements such as automated health-result monitoring, dietitian alerts, trend analysis, health-risk notifications, and diet plan review workflows.

Major Stakeholders/Users with specific roles:

- **Mr. Nuwan Perera** – Operations Manager
- **Ms. Ishara Jayasinghe** – Senior Dietitian
- **Mr. Chamara Fernando** – Medical Center Coordinator
- **Ms. Dilki Ranatunga** – Patient Relations Officer
- **Mr. Sahan Wickramasinghe** – Finance and Administration Executive

Core Functions

1. Patient Management

Description: manages patient information throughout their relationship with the healthcare center. It allows patients to register, maintain personal details, update medical information, and access their health records securely.

Sub-functions:

- Patient registration
- Login and profile management
- Medical history management
- Allergy and health condition records

2. Appointment Management

Description: handles the scheduling of consultations and health check-ups. It enables patients to book, reschedule, or cancel appointments while helping staff manage specialist schedules and reduce booking conflicts.

Sub-functions:

- Book appointments
- View available schedules
- Reschedule appointments
- Cancel appointments
- Appointment reminders

3. Health Check-up Management

Description: records and stores health assessment data, including vital signs, laboratory results, medical observations, and consultation notes. It provides a complete health history for future reference and follow-up care.

Sub-functions:

- Record vital signs and measurements
- Store test results
- Add doctor/dietitian notes
- View health history
- Manage follow-up recommendations

4. Diet Plan & Progress Management

Description: supports dietitians in creating personalized diet plans and monitoring patient progress. It tracks health indicators such as weight, BMI, and nutritional goals to evaluate the effectiveness of dietary recommendations.

Sub-functions:

- Create and update diet plans
- Assign diet plans to patients
- Manage meal schedules
- Track weight, BMI, and health goals
- Monitor patient progress

5. Communication & Notifications

Description: improves communication between patients and healthcare professionals. It provides messaging capabilities and sends automated reminders, alerts, and updates related to appointments, diet plans, and health activities.

Sub-functions:

- Patient–dietitian messaging
- Appointment notifications
- Diet plan update alerts
- Follow-up reminders
- Health result alerts

6. Billing & Reporting Management

Description: manages financial transactions and organizational reporting. It records payments, generates invoices, tracks financial activities, and produces reports that support management decision-making and business performance monitoring.

Sub-functions:

- Generate invoices
- Record payments
- Manage refunds
- Financial reports
- Patient and appointment reports
- Staff workload reports